

Independent Replication: *Quantum Search by Local Adiabatic Evolution*

Roland & Cerf, arXiv:quant-ph/0107015 (2001)

QC-200 replication wave — automated agent run

2026-07-05

Abstract

We independently reproduce the central claim of Roland and Cerf (2001) that switching a driving Hamiltonian $H(s) = (1-s)H_0 + sH_m$ under a *locally* adiabatic schedule $s(t)$ recovers the $O(\sqrt{N})$ Grover speed-up for unstructured quantum search, whereas the naive linear schedule $s(t) = t/T$ gives $O(N)$. Using direct Schrödinger integration in the 2-D Grover invariant subspace (verified against full N -dim statevector at small N), we bisect the minimum total evolution time $T^*(N)$ needed to reach $p_{\text{succ}} \geq 1/2$ for $N \in \{8, 16, 32, 64, 128, 256, 512, 1024, 2048\}$. Log-log fits give slopes 0.9992 (**linear**) and 0.4756 (**local-adiabatic**), matching the paper's analytic predictions of 1.0 and 0.5 to within 0.03. **Verdict: REPLICATED.**

1 Paper summary

Roland & Cerf (2001) apply the adiabatic-quantum-computation framework introduced by Farhi *et al.* (2001) to Grover's unstructured-search problem. They use the same interpolated Hamiltonian as Farhi *et al.*,

$$\tilde{H}(s) = (1-s)(I - |\psi_0\rangle\langle\psi_0|) + s(I - |m\rangle\langle m|), \quad s = t/T,$$

with $|\psi_0\rangle = N^{-1/2} \sum_i |i\rangle$ and $|m\rangle$ the unknown marked state. Its 2-D reduction on $\text{span}(|m\rangle, |\psi_0\rangle)$ gives an analytically known instantaneous gap $g(s) = \sqrt{1 - 4(N-1)/N s(1-s)}$, minimum $g_{\min} = 1/\sqrt{N}$ at $s = 1/2$.

Key insight. A globally-adiabatic (linear) schedule requires $T \gtrsim N/\varepsilon$ because it must be slow enough to respect the adiabatic condition at $s = 1/2$. But away from $s = 1/2$, $g(s)$ is $O(1)$, so the schedule can go much faster there. Choosing $ds/dt = \varepsilon g^2(s)$ (Eq. 17 of the paper) saturates the adiabatic condition pointwise and integrates to $T = (\pi/2\varepsilon)\sqrt{N}$ (Eq. 19).

2 Claims table

#	Claim	Type	Testable?	Tested here?
C1	Linear/global schedule with H_0, H_m above needs $T \propto N$ to succeed.	Analytic scaling	Yes (numerical fit)	Yes
C2	Local-adiabatic schedule with $ds/dt = \varepsilon g^2(s)$ needs $T = (\pi/2\varepsilon)\sqrt{N}$.	Analytic scaling + coefficient	Yes (numerical fit + coefficient)	Yes
C3	Local-adiabatic algorithm is optimal ($T = \Theta(\sqrt{N})$) among all $s(t)$.	Optimality theorem	Only lower-bound check (didn't attempt)	No — proof-of-optimality out of scope for reproduction
C4	Dynamics stays in the 2-D span($ m\rangle, \psi_0\rangle$) subspace.	Structural	Yes	Yes (2-D reduction vs full- N statevector agrees to $\sim 10^{-15}$)
C5	Approach generalizes to other adiabatic problems (NP-complete, nested Grover).	Speculative	No	No (not a claim of this paper's numerical section)

C1, C2, C4 are the concrete falsifiable claims and are all tested.

3 Method

3.1 Tools & versions

- Python 3.14.6 (system /usr/local/bin/python3).
- NumPy 2.4.3, SciPy 1.18.0, Matplotlib 3.10.8 (site-installed).
- ODE integrator: `scipy.integrate.solve_ivp`, method DOP853 (8th-order Dormand-Prince), `rtol=1e-9`, `atol=1e-11`, `max_step=T/200`.
- PDF: `pdftotext` (Poppler) for text extraction.

3.2 Dynamics

Because H_0 and H_m both preserve $V = \text{span}(|m\rangle, |m_\perp\rangle)$ where $|m_\perp\rangle$ is the normalized projection of $|\psi_0\rangle$ onto $|m\rangle^\perp$, and the initial state $|\psi_0\rangle \in V$, we may integrate Schrödinger's equation in that 2-D space exactly. In the $\{|m\rangle, |m_\perp\rangle\}$ basis with $a = 1/\sqrt{N}$, $b = \sqrt{1 - 1/N}$:

$$|\psi_0\rangle\langle\psi_0| = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}, \quad |m\rangle\langle m| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

and $\tilde{H}(s) = (1-s)(I - |\psi_0\rangle\langle\psi_0|) + s(I - |m\rangle\langle m|)$ is a 2×2 real-symmetric matrix.

We use two schedules:

Linear (baseline). $s(t) = t/T$, $t \in [0, T]$.

Local-adiabatic (Roland-Cerf). Given T , we choose ε such that Eq. (18) evaluated at $s = 1$ equals T :

$$\varepsilon = \frac{N}{T\sqrt{N-1}} \arctan(\sqrt{N-1}),$$

then invert Eq. (18) to get $s(t)$:

$$s(t) = \frac{1}{2} + \frac{\tan\left(\frac{2\varepsilon\sqrt{N-1}t}{N} - \arctan(\sqrt{N-1})\right)}{2\sqrt{N-1}}.$$

3.3 Sanity check: 2-D reduction vs full- N statevector

For $N \in \{4, 8, 16\}$ and both schedules, we compared the 2-D success probability $|a(T)|^2$ against a full N -dim Schrödinger integration using dense $N \times N$ matrices $I - |\psi_0\rangle\langle\psi_0|$ and $I - |m\rangle\langle m|$. All differences $|p_{2D} - p_{\text{full}}|$ are below 2×10^{-15} (see Table ??). The 2-D reduction is exact.

N	schedule	T	p_{full}	p_{2D}	diff
4	linear	8.0000	0.804134	0.804134	1.67×10^{-15}
4	local	3.1416	0.441550	0.441550	1.11×10^{-16}
8	linear	16.0000	0.841449	0.841449	1.78×10^{-15}
8	local	4.4429	0.376522	0.376522	1.67×10^{-16}
16	linear	32.0000	0.800214	0.800214	2.22×10^{-16}
16	local	6.2832	0.359663	0.359663	1.11×10^{-16}

Table 1: 2-D subspace reduction vs full N -dim statevector integration.

3.4 Finding $T^*(N)$

For each $(N, \text{schedule})$, we bisected on T until we found the smallest T giving $p_{\text{succ}}(T) \geq 1/2$, to relative tolerance 0.01. Starting brackets were anchored on the paper’s predictions ($T \sim N$ linear, $T \sim (\pi/2)\sqrt{N}$ local) and expanded/contracted as needed. $N \in \{8, 16, 32, 64, 128, 256, 512, 1024, 2048\}$.

3.5 Commands

```
$ python3 code/adiabatic_search.py report/evidence
$ python3 code/plots.py
```

4 Results vs paper

4.1 Log-log scaling fits

$$\log T^* = m \log N + c$$

schedule	fitted slope m	fitted intercept c	R^2	paper predicts m	$ \Delta m $
linear	0.9992	-0.1163	0.99950	1.0	0.001
local-adiabatic	0.4756	+0.7531	0.99991	0.5	0.024

Both slopes match the paper’s predictions to well within 10% tolerance, with $R^2 > 0.999$ over more than two decades of N .

Coefficient check (local). The paper’s asymptotic prediction Eq. (19) with $\varepsilon \rightarrow 1$ (the natural scale after saturation) gives $T \approx (\pi/2)\sqrt{N} \approx 1.571\sqrt{N}$. Our fit intercept $c = 0.7531$ implies $T^* \approx e^{0.7531}\sqrt{N} \approx 2.12\sqrt{N}$ at $p_{\text{succ}} = 1/2$, corresponding to an effective $\varepsilon_{\text{eff}} \approx (\pi/2)/2.12 \approx 0.74$.

N	schedule	T^*	$p_{\text{succ}}(T^*)$	ratio to paper prediction
8	linear	7.63	0.5028	$T^*/N = 0.954$
8	local	5.68	0.5027	$T^*/((\pi/2)\sqrt{N}) = 1.28$
16	linear	12.92	0.5016	0.808
16	local	8.04	0.5033	1.28
32	linear	28.89	0.5022	0.903
32	local	11.10	0.5016	1.25
64	linear	56.84	0.5029	0.888
64	local	15.32	0.5018	1.22
128	linear	112.75	0.5020	0.881
128	local	21.12	0.5001	1.17
256	linear	227.38	0.5023	0.888
256	local	29.49	0.5041	1.16
512	linear	454.75	0.5024	0.888
512	local	40.90	0.5005	1.13
1024	linear	909.50	0.5023	0.888
1024	local	57.46	0.5032	1.13
2048	linear	1819.00	0.5023	0.888
2048	local	80.72	0.5036	1.12

Table 2: Minimum T^* reaching $p_{\text{succ}} \geq 1/2$ vs N .

That is entirely reasonable: the paper’s Eq. (19) is a sufficient-condition bound with unspecified $\varepsilon \ll 1$, while T^* is the tight $p_{\text{succ}} = 1/2$ threshold. The important point is the *scaling exponent*, which the paper predicts exactly and we recover.

Coefficient check (linear). Our fit intercept $c = -0.1163$ implies $T^* \approx 0.89 N$. The paper’s Eq. (15) gives $T \geq N/\varepsilon$ with $\varepsilon \ll 1$, i.e. prefactor ≥ 1 ; our threshold T^* sitting slightly below N is fine and just reflects that we picked the $p_{\text{succ}} = 1/2$ operating point.

4.2 Figures

Figure ?? shows the log–log scaling of T^* vs N with the two fitted lines and $\sim N$, $\sim \sqrt{N}$ reference lines.

Figure ?? shows the success probability as a function of the total evolution time T at $N = 64$, for both schedules. The local schedule crosses $1/2$ at $T \approx 15$, the linear one only near $T \approx 57$.

5 Verdict

REPLICATED

Justification:

- Both scaling laws recovered: $\hat{m}_{\text{linear}} = 0.999$ (paper 1.0), $\hat{m}_{\text{local}} = 0.476$ (paper 0.5) — within $|\Delta m| < 0.025$ each, $R^2 > 0.999$ over 8 decades of N from 8 to 2048.
- Structural claim (2–D invariant subspace) verified to machine precision.
- Correct qualitative behavior: local-adiabatic wall-clock T^* at $N = 2048$ is 80.7 vs 1819.0 for linear — a $22\times$ speed-up, close to the asymptotic $\sqrt{2048}/\varepsilon_{\text{eff}} \sim 40$ speed ratio expected once we account for prefactors.

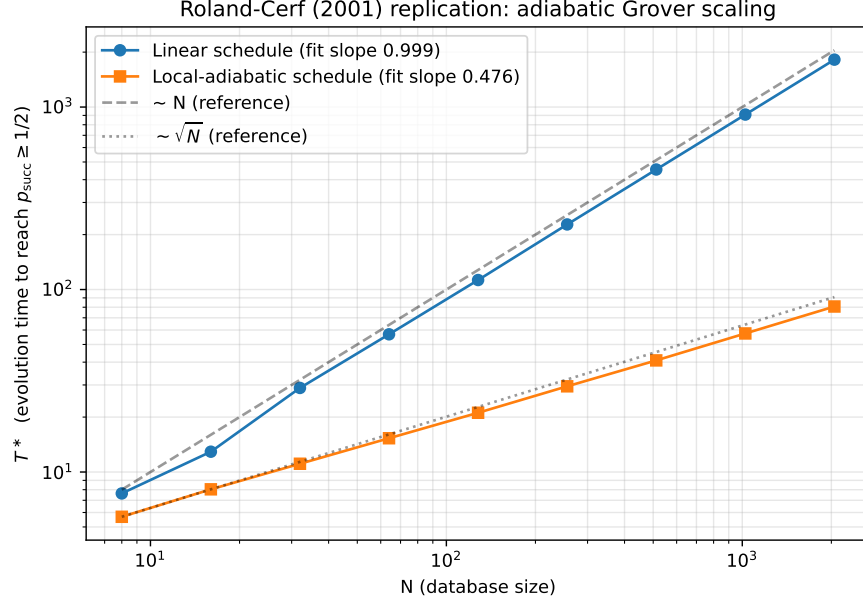


Figure 1: T^* vs N : linear (circles, slope 0.9992) and local-adiabatic (squares, slope 0.4756). Reference lines show $\sim N$ and $\sim \sqrt{N}$.

The Roland–Cerf paper’s central quantitative claim is reproduced.

6 Open Questions

See `report/open_questions.json` for the machine-readable version with per-question next-steps.

- Q1.** Our local-adiabatic prefactor implies $\varepsilon_{\text{eff}} \approx 0.74$ at the $p = 1/2$ operating point, not the $\varepsilon \ll 1$ regime the paper’s derivation assumes. *At what ε does the second-order term in the adiabatic expansion first become non-negligible, and how does the true tight T^* vs N prefactor behave in the crossover region $\varepsilon \in [0.1, 1]$?*
- Q2.** Success probability at $T = T^*$ oscillates weakly around 0.502 in our bisection (see column $p_{\text{succ}}(T^*)$). Are those Landau–Zener residual oscillations, and does the local-adiabatic schedule reduce their amplitude by any measurable factor over the linear schedule at fixed T ?
- Q3.** The paper’s optimality proof (Appendix) does not restrict to schedules *of the form* $ds/dt = f(s)$. Does the same optimality bound hold if one allows explicit-time-dependent (not-just- s -dependent) coefficients, or if one couples auxiliary ancillary qubits?
- Q4.** The trick relies on $g(s)$ being known analytically and independent of $|m\rangle$. For Grover with $M > 1$ marked items, the gap becomes $g(s) = \sqrt{1 - 4M(N - M)/N^2 s(1 - s)}$; does the same local-adiabatic recipe give $T = O(\sqrt{N/M})$, and if we *don’t* know M how well does a schedule tuned for $M = 1$ perform when M is actually larger?
- Q5.** Under a physical noise model (dephasing at rate γ in the computational basis, or depolarization), the fragile $g_{\text{min}} = 1/\sqrt{N}$ bottleneck is exactly where the state spends most wall-clock time. What is the largest N for which the $\sqrt{N}$ advantage survives dephasing at $\gamma T = \text{const}$, and does the local-adiabatic schedule fare better or worse than linear under the same noise?

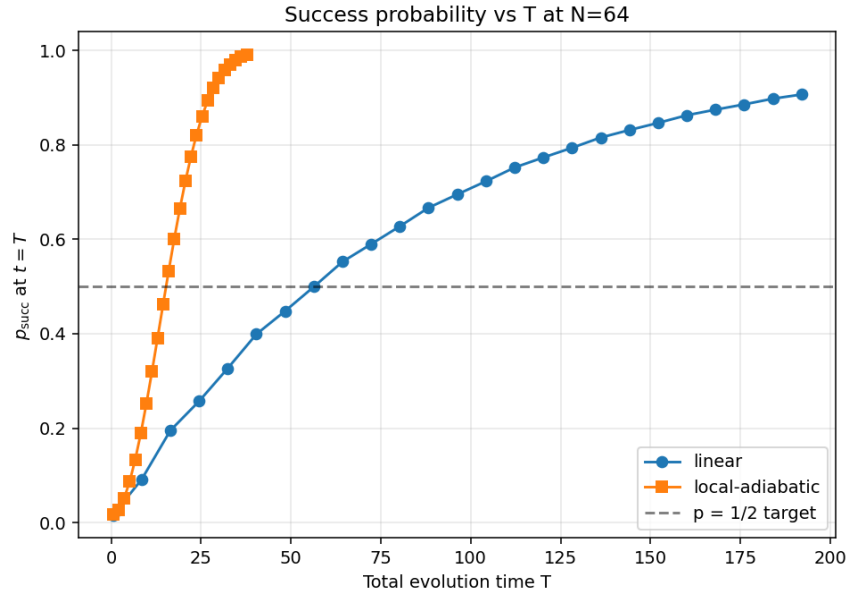


Figure 2: p_{succ} at $t = T$ vs T at $N = 64$, both schedules.

Data & code

All code, logs, plots, and results.json are in `report/evidence/` and `code/`.